

SAME PAGE, DIFFERENT BOOKS: EXTENDING REPRESENTATIONAL GAPS THEORY TO ENHANCE PERFORMANCE IN MULTITEAM SYSTEMS

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Multiteam systems are increasingly used by organizations, but are difficult to coordinate effectively. Building from theory on representational gaps, we explain why coordination between teams in multiteam systems can be hindered by inconsistencies that exist between them regarding the definition of shared problems. We argue that frame-of-reference training, an intervention that is theorized to reduce representational gaps, can facilitate coordination and subsequent performance in multiteam systems. We studied the effects of frame-of-reference training on multiteam system coordination and performance in 249 multiteam systems comprised of specialized teams. Supporting our expectations, we found that: (a) frame-of-reference training had a positive effect on multiteam system performance by enhancing between-team coordination; (b) within-team coordination improved the relationship between frame-of-reference training and between-team coordination, ultimately improving multiteam system performance; and (c) the patterns by which within-team coordination leveraged our intervention depended on teams' specific functions. Our findings extend representational gaps theory, highlight the interdependencies between team-level and system-level coordination, and demonstrate the utility of frame-of-reference training in a new setting.

In the modern workplace, many organizational challenges lie beyond the scope of any single team,

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requiring the use of networks of teams called *multiteam systems* (Zaccaro, Marks, & DeChurch, 2011). These hybrid organizational forms are comprised of at least two teams that exhibit interdependencies in their pursuit of shared goals (Mathieu, Marks, & Zaccaro, 2001). High-tech companies use multiteam systems to develop and launch products; pharmaceutical and biotech firms employ them to research and develop new drugs; and they are frequently used in manufacturing (e.g., by Boeing and Nissan; Lanaj, Hollenbeck, Ilgen, Barnes, & Harmon, 2013; Marks & Luvison, 2012; O'Leary, Woolley, & Mortensen, 2012), health care (Mathieu et al., 2001), and emergency response (DeChurch, Burke, Shuffler, Lyons, Doty, & Salas, 2011). The emerging literature is clear that, in addition to *within-team coordination* (i.e., integrating the actions of a specific team's

members; Marks, Mathieu, & Zaccaro, 2001; Thompson, 1967), *between-team coordination* (i.e., integrating the actions of functionally specialized component teams) is a critical process that enables multiteam systems to leverage a wide range of members' skills, backgrounds, and functions while flexibly directing the actions of their specialized teams toward a common objective (Davison, Hollenbeck, Barnes, Slesman, & Ilgen, 2012; DeChurch & Marks, 2006; Lanaj et al., 2013; Marks, DeChurch, Mathieu, Panzer, & Alonso, 2005).

Despite its critical role in multiteam system performance (i.e., effectiveness in achieving shared goals; Campbell, 1990; Mathieu et al., 2001), current theorizing has not fully explained what facilitates between-team coordination (DeChurch & Marks, 2006). What little is known has focused largely on the antecedents of leadership functions and structural design (e.g., centralization; DeChurch & Marks, 2006; Lanaj et al., 2013). However, other factors are likely important. The broader literature on coordination suggests that one particularly important way of enhancing coordination is to reduce *representational gaps*, or inconsistencies between units regarding how shared problems are conceptualized (Bruns, 2013; Cronin & Weingart, 2007; DeChurch & Mesmer-Magnus, 2010; Dougherty, 1992; Okhuysen & Bechky, 2009; Rentsch & Staniewicz, 2012). One illustration of the difficulties that representational gaps can cause for multiteam systems is that they frequently plagued coordination efforts in the multiteam system used to topple Osama bin Laden during Operation Neptune Spear, leading to costly human and material losses as well as severe delays to the completion of mission objectives (Allison, 2012; Bergen, 2012).

Theory on whether representational gaps can enhance coordination has not accounted for the additional complexities found in multiteam systems, such as the specialized subgoals that are uniquely pursued by each component team (Cronin & Weingart, 2007; Okhuysen & Bechky, 2009). Much of the work related to reducing representational gaps has made strong assumptions about high levels of interdependencies between all members. These assumptions may not typically hold in the functionally differentiated work found in multiteam systems, where it is better if not all members interact across team boundaries (Davison et al., 2012; cf. Galbraith, 1973). The work that has accounted for weaker interdependencies between members in functionally specialized units has largely focused on translating between them (e.g., Barley, 1996;

Davison et al., 2012; Gittell, 2002), but has not addressed how to reduce the underlying inconsistencies (i.e., representational gaps) that require translation. Even if theory explained why and when reducing representational gaps enhances between-team coordination, *how* to ensure this occurs in practice is not straightforward (Cronin & Weingart, 2007).

To complicate matters further, although there is evidence that multiteam system performance outcomes are directly impacted by the efforts of each component team (e.g., DeChurch & Marks, 2006; Marks et al., 2005), it is not yet clear whether or not between-team coordination depends on the quality of team-level processes such as within-team coordination. This shortcoming overlooks the hierarchical and interdependent nature of multiteam systems and has serious implications for designing and assessing interventions to enhance between-team coordination. If between-team coordination is largely independent of team-level processes, then efforts to enhance between-team coordination can safely ignore, for example, within-team coordination. However, if between-team coordination depends on the quality of team processes, then efforts to enhance between-team coordination can only be understood by accounting for within-team coordination. This latter scenario may be even more complex given evidence that the extent to which within-team coordination is important to system-level functioning depends on a team's specialized subgoals and associated tasks (Davison et al., 2012).

The purpose of this paper is to develop theory that explains what might enhance between-team coordination and subsequent multiteam system performance, and when this is most likely to occur. We build from information processing theory and extend representational gaps theorizing to explain why reducing representational gaps via frame-of-reference training—an intervention originating in the performance appraisal literature (Bernardin & Buckley, 1981)—can enhance between-team coordination and subsequent multiteam system performance. We thus clarify an important antecedent to between-team coordination. We also contribute to theory on the relationship between the hierarchical structure and interdependent functions of multiteam systems by articulating why between-team coordination (a system-level process) is dependent upon within-team coordination (a team-level process) (cf. Sinha & Van de Ven, 2005; Thompson, 1967). In doing so, we further

explain why within-team coordination enhances the effect of frame-of-reference training in multi-team systems, and why the pattern of this relationship depends on the function of a given component team. We test our hypotheses through a field experiment in a context in which representational gaps are embedded.

THEORY AND HYPOTHESES

Multiteam systems are comprised of component teams that exhibit interdependencies in their inputs (i.e., shared inputs required to accomplish their goals), processes (i.e., teams must interact with one another to accomplish their goals), and outcomes (i.e., team consequences depend on those of others) with at least one other team in the system (Mathieu et al., 2001). These interdependencies set multiteam systems apart from other organizational forms such as task forces, subassemblies, or matrix organizations. Another hallmark of multiteam systems is the presence of a goal hierarchy, comprised of an overarching shared goal at the multiteam system level that is supported by unique subgoals at the team and member levels (Mathieu et al., 2001; Zaccaro et al., 2011). As teams' efforts are integrated, the accomplishment of team-level goals can facilitate multiteam system-level goal accomplishment (Marks et al., 2005).

Owing to their unique environmental demands and subgoals, some teams' functions are more important to a multiteam system's functioning while other teams' functions, although still valuable, are less important. Prior work has established that when a particular team (a) is more central to a multiteam system's workflow, (b) encounters more of a multiteam system's shared problems, and (c) is exposed to more multiteam system tasks, it is more core, or crucial, whereas a team that is lower with respect to these criteria is non-core (Davison et al., 2012; Humphrey, Morgeson, & Mannor, 2009; Summers, Humphrey, & Ferris, 2012). In the multiteam system literature, core teams have been referred to as *point teams*, whereas non-core teams have been referred to as *support teams* (Davison et al., 2012). For consistency, we adopt these labels.

Due to the interdependencies between component teams and the presence of a shared goal hierarchy within multiteam systems, both within- and between-team coordination are crucial to multiteam system performance (Mathieu et al., 2001). *Within-team coordination* involves

information exchange and mutual adjustment of members' actions, and permits the unique capabilities of each team member to contribute to a team's specialized tasks and thus achieve their goals (Marks et al., 2001). High levels of within-team coordination enable team-level goal accomplishment (LePine, Piccolo, Jackson, Mathieu, & Saul, 2008; Mathieu et al., 2001), which subsequently contributes to goal accomplishment further up the multiteam system goal hierarchy (DeChurch & Marks, 2006). However, given the interdependent nature of component teams, multi-team system-level goals cannot all be achieved as simply an additive compilation of team-level efforts. *Between-team coordination* is the mechanism by which the efforts of multiple specialized teams are integrated to service higher-order goals, and is thus a central process in multiteam systems (Mathieu et al., 2001). Between-team coordination entails processing information obtained from multiple component teams, and synchronizing, integrating, and aligning their interdependent actions (Davison et al., 2012; DeChurch & Marks, 2006; cf. Hoegl, Weinkauff, & Gemuenden, 2004; Okhuysen & Bechky, 2009).

Information processing theory suggests that reducing uncertainty between teams regarding the meaning and importance of information that is shared between them may facilitate between-team coordination. Viewed through the lens of information processing theory, organizations are open systems that collect, gather, and process information—permitting them to respond to both internal and external sources of uncertainty (Tushman & Nadler, 1978)—and are comprised of interdependent but specialized subunits. Information processing theory suggests that two important sources of uncertainty are (1) the extent to which interdependent subunits (e.g., teams) rely on information that is shared between them, and (2) the characteristics of each subunit's (e.g., team's) task and environment.

We focus on two factors that each relate to one of these sources of uncertainty, and explain how reducing both factors can enhance between-team coordination. First, we argue that efforts targeted at reducing representational gaps can reduce uncertainty between teams regarding the meaning and importance of problems that are shared between them. Second, we argue that within-team coordination can reduce uncertainty regarding characteristics of each team's task and environment. We also articulate how these two factors interact to

strengthen between-team coordination, and, ultimately, multiteam system performance.

Extending Representational Gaps Theorizing: Enhancing Between-Team Coordination via Frame-of-Reference Training

To better explain why inconsistencies between teams regarding how shared problems are cognitively represented can inhibit between-team coordination (Bechky, 2003; Cronin & Weingart, 2007; Dougherty, 1992; Okhuysen & Bechky, 2009; Rico, Sánchez-Manzanares, Gil, & Gibson, 2008), we build from representational gaps theorizing. Although originally developed to explain why actions may be poorly integrated between specialized members within stand-alone teams, we extend representational gaps theory to multiteam systems in which actions may be poorly integrated between specialized teams. Multiteam systems ideally possess a cognitive structure that consists of a *goal hierarchy* (i.e., shared objectives and their relative importance), *assumptions* (i.e., preferences and constraints that may partially overlap between component teams), *elements* (i.e., specific languages used to deal with their specific tasks), and *operators* (i.e., rules regarding the meaning and utility of information; Cronin & Weingart, 2007). Differences in these cognitive structures between teams result in *representational gaps*, which are between-unit inconsistencies regarding the conceptualization of a shared problem (Cronin & Weingart, 2007).

Current theorizing on representational gaps assumes that, to be effective, any actions members take at a given time must be oriented toward the same objectives. What is missing is the recognition that goal hierarchies can include nested subgoals that are uniquely dispersed between members, or, in the case of multiteam systems, between teams. The presence of different subgoals leads teams to each work on distinctive objectives (rather than the same one(s) suggested by present theorizing). For example, in Operation Neptune Spear, a CIA support team frequently pursued the subgoal of determining the identity of bin Laden's close aids, while a special forces point team pursued the subgoal of tracking the location of known operatives. Specialized actions such as these correspond with very different systems for understanding, communicating about, and assessing teams' specialized subgoals (i.e., different elements and operators), likely increasing uncertainty around how any particular

team's actions can be incorporated into the broader system.

Given the fundamental importance of each component team pursuing distinct yet complementary subgoals in multiteam systems (Mathieu et al., 2001), if between-team coordination is to occur, then idiosyncratic operators and elements must be reduced so that information from one team can be understood and used by other teams or the system as a whole (Cronin & Weingart, 2007; Mathieu et al., 2001; Rentsch & Staniewicz, 2012; cf. Cronin, Bezrukova, Weingart, & Tinsley, 2011; Dahlin, Weingart, & Hinds, 2005). Generally, this can be accomplished by providing a system that identifies elements of and relationships between task components, enabling a shared understanding that facilitates translation between teams. Without such a system, "unshared elements across [component teams] present the same problem as unshared language—interference with the transmission of meaning both explicitly and implicitly" (Cronin & Weingart, 2007: 767).

To provide a specific approach that might enhance between-team coordination by aiming to reduce representational gaps, we turn to the performance appraisal literature. *Frame-of-reference training* has long been recognized as an effective means of reducing idiosyncratic perceptions between individuals by providing standardized rating scales that can guide individuals' (e.g., raters of performance) actions (Athey & McIntyre, 1987; Dierdorff, Surface, & Brown, 2010; Gorman & Rentsch, 2009; Woehr, 1994). When individuals can compare their decision to a clear standard, they make more accurate and reliable judgments (Day & Sulsky, 1995; Noonan & Sulsky, 2001). By improving the internal consistency of individuals, consistency across individuals also improves (Schleicher & Day, 1998). Even providing loose standards can improve group decision accuracy in non-performance-appraisal settings (Laughlin, Bonner, Miner, & Carnevale, 1999).

Frame-of-reference training likely enhances between-team coordination by reducing idiosyncrasies between teams with regard to operators and elements (i.e., the languages and rules used). The shared standards provided by frame-of-reference training permit different teams to be "on the same page" in terms of how to conceptualize and communicate their work with regards to the broader multiteam system-level operating environment, enabling a clear translation from team-specific environmental features to assessments of these features

that can be clearly communicated between teams. As a result, frame-of-reference training should reduce idiosyncratic and subjective conceptualizations of teams' different environments, task demands, and interdependent actions that otherwise make coordinating actions between teams difficult.

For the same reasons, frame-of-reference training also likely reduces the effort required to interpret teams' actions and decreases the number of misunderstandings between teams—and resultant errors or inefficiencies—during between-team coordination. Initial theorizing suggests that reducing representational gaps decreases the likelihood of system-level actions being at odds with any specific teams' efforts due to a better understanding of how actions oriented toward teams' subgoals can be unified toward a common goal. It is possible that, by making component teams' actions easier to understand and relate to one another, frame-of-reference training might make it easier to detect conflicting actions that would otherwise have not been apparent, thus disrupting coordination efforts while that conflict is managed. This may temporarily interrupt between-team coordination. However, becoming aware of such conflicting actions makes it more likely that conflict can be resolved, thus permitting system-level coordination and performance to improve in the long run more than if members unknowingly continued to counteract one another. Other important factors, such as psychological safety, trust, and relational commitment, may be important to ensure that differing viewpoints are integrated into multiteam system processes (e.g., Bradley, Postlethwaite, Klotz, Hamdani, & Brown, 2012; Edmondson, 1999; Lawler & Yoon, 1996), but these factors are less likely to be helpful if, in the absence of frame-of-reference training, members struggle to understand one another.

By enhancing between-team coordination, we expect that frame-of-reference training will also indirectly improve multiteam system performance. Research has shown that between-team coordination positively relates to multiteam system performance (Davison et al., 2012; DeChurch & Marks, 2006; Lanaj et al., 2013). When between-team coordination is high, component teams' subgoals are better aligned and multiteam system-level actions can contribute above and beyond teams' actions to accomplish system-level goals (Mathieu et al., 2001).

Hypothesis 1. The positive relationship between frame-of-reference training and multiteam system performance is mediated by between-team coordination.

Relating Within-Team and Between-Team Coordination

Relying on information processing theory as a framework, the information processing requirements associated with coordinating actions throughout the entire multiteam system (i.e., between-team coordination) can be reduced if uncertainty can be reduced at the sub-unit level (i.e., in component teams). Within-team coordination can reduce the uncertainty arising from component teams' task environments, which is one of the major sources of uncertainty that can impede between-team coordination. When a team's members' actions are better aligned and information is exchanged more readily between them (i.e., within-team coordination is high), information about the teams' operating environment is more likely to be accounted for. That is, team processes and outcomes are more likely to account for local environmental demands, which tend to be dynamic (Mathieu et al., 2001; Zaccaro et al., 2011). When these processes or outcomes are used as inputs by the system, there is less uncertainty about any given teams' local environmental demands. This reduced uncertainty enables those engaged in between-team coordination to exert less effort scanning the environment and interpreting teams' actions—that is, it reduces information-processing demands—and enables them to devote more attention toward ensuring that teams' actions are aligned appropriately toward higher-level goals (cf. Schmidt & DeShon, 2007).

Hypothesis 2. Within-team coordination of (a) the point team and (b) the support team positively relates to between-team coordination.

Improving the Within-Team and Between-Team Coordination Relationship: Accounting for Team Functions

Although information processing theory holds that reducing uncertainty from different sources should additively enhance information-processing capacities of the system, we suggest that reducing uncertainty from multiple sources can have an interactive effect in benefiting between-team coordination. Even if frame-of-reference training reduces uncertainty around information shared between teams by mitigating idiosyncratic operators or elements, the extent to which the accompanying shared understanding and common language benefit between-team coordination

likely depends on whether or not component teams reduce uncertainty with respect to their own tasks and environments. Given fundamental differences in how component teams' subgoals relate to the broader system, the pattern in which this interactive relationship occurs likely differs across component team functions (Davison et al., 2012).

We expect that the positive relationship between frame-of-reference training and between-team coordination is *enhanced* by within-team coordination in point teams. As a result of the characteristics that make point teams more core, their subgoals are more proximal to performance of the system, and, thus, their actions are a necessary condition for between-team coordination. During between-team coordination, system-level actions are more dependent on ensuring that the actions of support teams contribute to the actions of point teams than vice versa (cf. Davison et al., 2012; Humphrey et al., 2009). When within-team coordination in point teams is higher, uncertainty around the system's most central tasks will have been reduced, providing a more straightforward understanding of how to incorporate supplementary actions into and around those tasks, and reducing potential errors that such uncertainty may otherwise cause. Accordingly, the shared understanding and common language made possible by frame-of-reference training permits supportive actions to be even better integrated with the actions taken by point teams toward their more proximal subgoals when within-team coordination for point teams is higher. However, when within-team coordination in point teams is lower, greater levels of uncertainty likely remain around tasks that are central to the system, making it more difficult to integrate actions from other teams that relate to those central tasks regardless of any benefits provided by frame-of-reference training.

Hypothesis 3. The strength of the mediated relationship between frame-of-reference training via between-team coordination and multiteam system performance will vary depending on the level of within-team coordination of point teams, such that the indirect effect of frame-of-reference training via between-team coordination on multiteam system performance will be stronger when within-team coordination of point teams is higher.

We also expect that within-team coordination in support teams moderates the relationship between

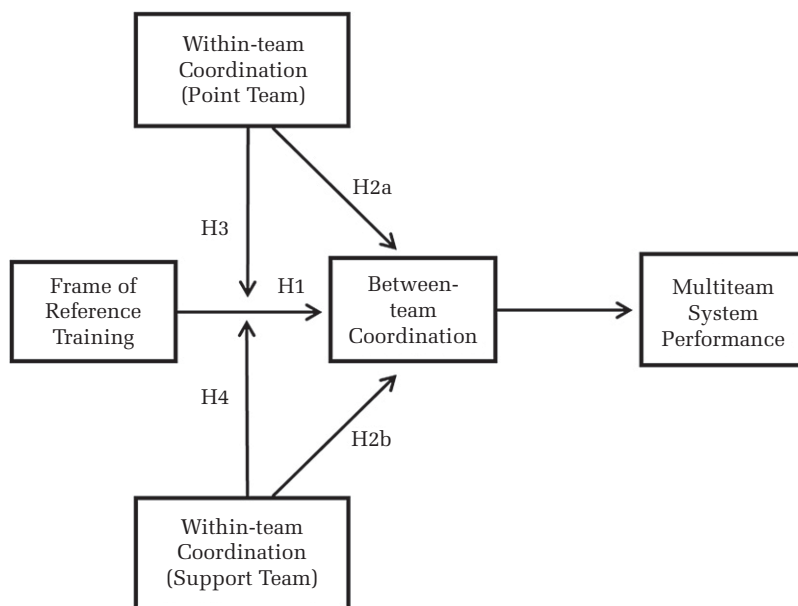
frame-of-reference training and between-team coordination, but in a different pattern. Specifically, within-team coordination in support teams likely has a *compensatory* effect on the relationship between frame-of-reference training and between-team coordination, such that it is strongest when within-team coordination is low. We still expect that high levels of within-team coordination in support teams will benefit the relationship between frame-of-reference training and between-team coordination, but that the benefit will be far less pronounced than when within-team coordination in support teams is low.

Although the auxiliary benefits provided to the system by support teams is critical, multiteam system goal hierarchies are such that point teams and the overall system can (and often must) take actions regardless of the quality of support team's actions. When within-team coordination in support teams is low, the presence of frame-of-reference training is particularly important to between-team coordination. In this case, even poorly coordinated actions passed on from support teams can be put to use, as frame-of-reference training permits an understanding of how at least some of the coordinated actions of the support team might be useful at the system level (i.e., what little contribution support teams provide can be understood and integrated). On the other hand, when within-team coordination in support teams is low and frame-of-reference training is not present, it is particularly difficult to understand how the support teams' coordinated actions might be useful during between-team coordination. In this case, support teams' actions are both of lower quality (i.e., there is greater uncertainty regarding their environment and tasks) and are more difficult to interpret or make sense of when applied to the system, thus providing little additional value to between-team coordination.

Hypothesis 4. The strength of the mediated relationship between frame-of-reference training via between-team coordination and team performance will vary depending on the level of within-team coordination of support teams, such that the indirect effect of frame-of-reference training via between-team coordination on multiteam system performance will be stronger when within-team coordination of support teams is lower.

To summarize, Figure 1 illustrates the conceptual model captured by our four hypotheses and tested in

FIGURE 1
Hypothesized Model



this study. At the core of this model is the notion that frame-of-reference training has an effect on multiteam system performance by enhancing between-team coordination, and that the value of frame-of-reference training to the multiteam system depends on levels of within-team coordination in a pattern that is dependent on a given team's function.

METHODS

Sample and Task

The sample was taken from a larger, multiyear program of research¹ sponsored by the United States Air Force (USAF) and a large research-oriented

university. The research participants were USAF captains attending a five-week leadership development course. These officers had a wide range of occupational backgrounds and experiences, making the sample diverse in skills and abilities. A total of 3,486 officers were randomly assigned to one of 249 14-person multiteam systems for the entire five-week course, such that each multiteam system was comparable with respect to members' ability, gender, experience, age, and vocational specialty. Performance on the task comprised part of the overall evaluation of the participants in the leadership course, with ramifications for later job assignments and promotions that likely motivated them to perform well. The task was also similar to work that many officers performed during normal duties, and thus provided a level of authenticity lacking in many lab experiments.

For our task, we used a leadership development simulation (see Appendix A for a screenshot) conceived and developed specifically for this research and training exercise. The simulation task involved directing a set of remotely piloted aircraft (RPA) to search for, identify, and engage enemy targets on a 16×16 grid over a series of 10 rounds. The task required both coordination and memory-based reasoning under time pressure in order to find and engage targets. The overarching goal of the

¹ Given this multiyear program, our sample has some overlap with prior studies. Of the 249 multiteam systems in our sample, 90 overlap with those reported in Lanaj et al. (2013), and 70 overlap with those reported in Davison et al. (2012). We introduced a novel manipulation in our study, received by 119 multiteam systems. Those receiving our manipulation had no overlap with the sample reported in Davison et al. (2012), but included 14 teams reported in Lanaj et al. (2013). However, our manipulation was not within the theoretical scope of Lanaj et al.'s (2013) paper.

multiteam system was to maximize points, which were awarded for engaging enemy targets and deducted for losing RPAs or bases to enemy strikes, and required all component teams to work together interdependently.

The primary simulation mechanism used by multiteam systems to coordinate their actions was the “common operational picture” (COP), a map of the 16×16 game grid on which target indicators could be placed that was displayed on one wall of the room in view of all team members. When component teams believed that they had located a target, they notified the leadership team, which placed an indicator on the COP for that target. Of particular importance to our intervention, this target indicator could also be assigned a confidence rating, from “0” (no confidence) to “4” (ground truth), indicating the certainty of a target’s location.

Multiteam System Component Team Functions

Participants were assigned to specific roles within their 14-person multiteam system. First, they were assigned to either a point team (responsible for operational assets), a support team (responsible for intelligence assets), or a leadership team (responsible for updating the COP and overseeing system actions; see Appendix B for how members were divided across component teams). The component teams were designed to be functional in nature, with roles defined by the job responsibilities of each member. The point team was responsible for controlling the RPAs that the team used to engage targets and protect assets, and had to work interdependently to do so. The team consisted of a director and assistant director, who were the channels through which the point team received instructions from the leadership team; they also provided more proximal leadership functions to the point team. The other four members of the point team were each assigned four RPAs of a unique type: “strike” (for attacking opportunity targets), “escort” (for protecting missions from threats), “in-fo” (for collecting information on targets), and “re-fuel” (allowing for missions into deeper territory).

Similarly, the support team had a director and assistant director to provide leadership, as well as four members who were assigned a particular type of intelligence asset: “visual,” “human,” “comm” (communications), and “allied.” These assets differed in where on the game grid they provided accurate information, but otherwise provided intelligence on the square to which they were

deployed. Together, these members worked interdependently to identify targets. For example, to deploy two different types of intelligence asset to the same point on the grid so that their information could be verified against one another, it was necessary for two different members of the support team to work together to send out their unique assets.

A mission commander and vice commander, who had ultimate responsibility for the decisions of the multiteam system, were joined by the director and assistant director from the point and support teams to form the leadership team. This formalized the boundary-spanning role of the component team leaders, and enabled them to disseminate direction to the entire system. The leadership team maintained the COP, coordinated information between the component teams, made alterations to the missions forwarded by the component teams before they were finalized, and assisted component teams in performing their responsibilities.

Each role within each component team had unique responsibilities, and each component team had unique subgoals and tasks that were not duplicated elsewhere in the multiteam system (e.g., point team members could not control assets of the support team, and vice versa, and actions of the point team served different purposes than those of the support team). For example, in order for the multiteam system to send intelligence assets and RPAs to the same location on the game grid, the point team and the support team had to coordinate their activities with one another. The distinct differences between teams in terms of their subgoals, tasks, and the methods and terminologies they used to accomplish these tasks, as well as time constraints that prevented teams from becoming acquainted with the details of these differences, served to build significant representational gaps into the context of the study.

Training

Before participating in the simulation task, all multiteam system members—regardless of their experimental condition—were shown a 45-minute slide presentation in lecture format on the basic rules and functionality of the simulation (i.e., target information, rules, roles of team members, and other basic task knowledge). Afterward, each multiteam system was led through a 15-minute hands-on training session, in which members performed basic tasks in the simulation under the guidance of an instructor. During training, participants were given

the opportunity to ask questions about the simulation, and afterward were provided feedback on their performance. After training but prior to engaging in the task, for each multiteam system, all members of all component teams participated in synchronous planning to determine their strategies. Thus, all multiteam systems received the same basic training and had identical opportunities to communicate and plan prior to the task, with the exception of the details provided in the experimental manipulation, explained below.

Frame-of-Reference Training Manipulation

The experimental condition consisted of providing frame-of-reference training, which consisted of a brief modular addition to the broader task training described above. The control condition received no additional training. No team or multiteam system differences were used to determine inclusion in either condition. Given the critical role of the target indicators and their associated confidence ratings on the COP for performance, the frame-of-reference training was focused specifically on providing clear performance standards that could guide the accuracy with which members provided confidence ratings of the presence of specific targets on the COP. The criteria for confidence ratings provided a standardized language for translating judgments based on idiosyncratic, functionally specific factors into a generically interpretable format easily understood by individuals who may not share the same area of expertise. Similar to Laughlin et al. (1999), each decision made by the subjects in the present research had a definitive answer (i.e., when rating confidence of the presence of a target, that target was either really present or it was not). Prior research on frame-of-reference training has focused on improving decision accuracy, but has not directly aligned confidence with accuracy. Our frame-of-reference training is unique because it directly addresses how to most appropriately express confidence in a decision based on the criteria used to make that decision.

Our training followed the general recommendations provided in the literature. Thus, we included fine-grained behavioral criteria that were paired with specific ratings of confidence to be provided on the COP (Cardy & Keefe, 1994; Laughlin et al., 1999). Normative confidence-rating decisions were linked to every combination of observations of targets on the game grid in order to provide a useful and efficient script schema, summarized in a handout (see

Appendix C). We also provided verbal training accompanied by slides on how and when to utilize the performance criteria. Although the training details were specific to this unique task, the process of converting idiosyncratic, functionally specific factors into a generically interpretable format, easily understood by individuals who may not share the same area of expertise, via the construct of calibrated confidence levels is generic and could be applied to many other different tasks.

The verbal portion of frame-of-reference training was embedded in the broader training on how to engage in the task, and thus all members were present, but it was directed at the leadership team. The training material handouts were only provided to the vice commander and assistant directors (i.e., the three members of the leadership team who dealt most directly with updating the COP). These members were free to disseminate this information throughout the multiteam system so that all members might be aware of how to incorporate the training material. Multiteam systems in both conditions were equally informed as to the mechanics and importance of using confidence ratings, but those in the frame-of-reference training condition were given specific guidelines on performance criteria (i.e., what criterion should be used to indicate a particular level of confidence). Frame-of-reference training took approximately three additional minutes, during which time multiteam systems in the control condition were instructed that they were responsible for formulating criteria on their own. No element of this intervention was oriented toward drawing attention to or improving coordination of any other team process.

Measures

Frame-of-reference training. Of 249 multiteam systems, 119 received frame-of-reference training. Trained teams were coded "1"; untrained, "0."

Within-team coordination. We used separate variables to capture within-team coordination in each of the two specialized component teams. Within-team coordination in point teams was measured by the number of missions protected by escort RPAs (see Davison et al., 2012). This action required two or more members to coordinate and send escort RPAs to the same location as their other RPAs. Given that there were 256 squares on the grid and only four escort RPAs available per turn, the chances of this happening by accident were small. Coordinating these missions did not require the input

TABLE 1
Descriptive Statistics and Correlations

Variable	Mean	SD	1	2	3	4	5
1. Frame-of-reference training	.48	.50	—				
2. Within-team coordination (point team)	85	12.77	.04	—			
3. Within-team coordination (support team)	240	39.36	.11	.11	—		
4. Between-team coordination	51	21.46	.16*	.13*	.09	—	
5. Multiteam system performance	157	46.14	.08	.25*	.51*	.20*	—

Notes: $n = 249$; observations are multiteam systems of 14 members each.

* $p < .05$; two-tailed tests

of any members from other teams, making it an entirely within-team variable.

Within-team coordination in support teams was measured by the number of times a member deployed an intelligence asset to a square on the game grid where that asset had high accuracy. This measurement has been used in past studies as a measure of coordinated behavior within the support team (Davison et al., 2012). To effectively deploy these intelligence assets, the support team worked together to determine where the four different types of intelligence assets were accurate, and deployed those assets only where they provided accurate information. This process required the support group to internally coordinate their activities.

Between-team coordination. We measured between-team coordination behaviorally as the number of missions sent to targets on the COP with a confidence rating indicated (i.e., missions sent to targets with ratings of “1” through “4”; missions with confidence ratings of “0” were excluded; see Appendix C). Missions flown to targets indicated on the COP could only occur if the leadership team coordinated between component teams to ensure the point team knew where targets identified by the support team were located. In line with our conceptual definition of between-team coordination as the process by which functionally specialized actions of component teams are integrated, this measure captures interdependent and integrated system-level actions that required input from the support team (i.e., placing target ratings on COP), point team (i.e., sending missions to specific targets), and the leadership team (i.e., coordinating between teams; approving final missions; cf. Davison et al., 2012; Lanaj et al., 2013).

Multiteam systems that did not effectively engage in between-team coordination tended to engage unrated targets that were detected by the point team, and then immediately engaged by the point team.

This practice was not effective, as it precluded (a) deriving any additional value from the support team’s efforts to verify the accuracy of these targets and (b) the opportunity to see the larger picture of enemy deployment that was available from the COP being developed. Multiteam systems that were low in between-team coordination tended to have point teams that were largely working alone without their efforts being coordinated with the intelligence team or the leadership team.

Multiteam system performance.² Multiteam system performance was measured by the total points earned, which were gained by engaging targets and lost by the destruction of RPAs and/or bases by the enemy. This reflects our conceptualization of multiteam system performance as the effectiveness in achieving shared goals (Campbell, 1990), as the explicitly stated goal of all multiteam systems was to maximize total points earned.

RESULTS

Table 1 reports descriptive statistics, including means, standard deviations, and correlations. Given that our predictions were directional in nature, with the exception of Table 1, all reported analyses were obtained via one-tailed significance tests. Table 2 presents the hierarchical moderated regression analyses testing Hypotheses 1 and 2. To test Hypothesis 1, we examined regression equations and ran mediation analyses in the statistical modeling program Mplus 6.1 (Muthén & Muthén, 2010), in line with recommendations provided by Preacher, Rucker, and Hays (2007) and Edwards and Lambert (2007). Frame-of-reference training positively related to between-team coordination ($\beta = .16$; $p < .05$;

² This same measure of multiteam system performance was used in Lanaj et al. (2013), but no other variables were shared.

TABLE 2
Hierarchical Moderated Regression Analysis—Between-Team Coordination as Dependent Variable

	Model 1	Model 2	Model 3	Model 4	Model 5
Frame-of-reference training	.16*	.15*	.08 [†]	.09 [†]	.09 [†]
Within-team coordination (point team)		.12*	−0.12 [†]	−0.08	−0.08
Within-team coordination (support team)		0.06	.17*	.17*	.17*
Within-team coordination (point team) × Frame-of-reference training			.34*	.29*	.30*
Within-team coordination (support team) × Frame-of-reference training			−.15*	−.17*	−.17*
Within-team coordination (point team) × Within-team coordination (support team)				.13*	0.11 [†]
Within-team coordination (point team) × Within-team coordination (support team) × Frame-of-reference training					0.03
Total R^2	.03*	.04*	.11*	.12*	0.12
ΔR^2		.01*	.07*	.08*	0.08

Notes: $n = 249$; observations are multiteam systems of 14 members each; standardized betas are reported.

* $p < .05$

[†] $p < .10$; one-tailed tests

see Table 2). Controlling for frame-of-reference training and within-team coordination, between-team coordination positively related to multiteam system performance ($\beta = .18$; $p < .05$; see Table 3). Using estimates obtained from the test of mediation, a Sobel test indicated that, in line with our expectations, there was a positive indirect effect of frame-of-reference training on multiteam system performance, mediated by between-team coordination (2.28, $p < .05$).

After accounting for frame-of-reference training, within-team coordination of the point team demonstrated a significant positive relationship with between-team coordination ($\beta = .12$; $p < .05$), but within-team coordination of the support team did not demonstrate a significant positive relationship with between-team coordination ($\beta = .06$; $p = .17$; see Table 2). Thus, Hypothesis 2a, but not Hypothesis 2b,

was supported. The average level of within-team coordination across both point teams and support teams significantly and positively predicted between-team coordination ($\beta = .09$; $p < .05$).

To test Hypotheses 3 and 4, we first included the respective interaction terms into the regression model (see Table 2). After accounting for the main effects of frame-of-reference training and within-team coordination in support teams, within-team coordination in point teams increased the relationship between frame-of-reference training and between-team coordination ($\beta = .34$; $p < .05$; see Figure 2 for the form of this interaction, which uses values of ± 1 SD for within-team coordination). The nature of this interaction suggests that frame-of-reference training is most beneficial when within-team coordination for point teams is high. The interaction between within-team coordination in

TABLE 3
Hierarchical Moderated Regression Analysis—Multiteam System Performance as Dependent Variable

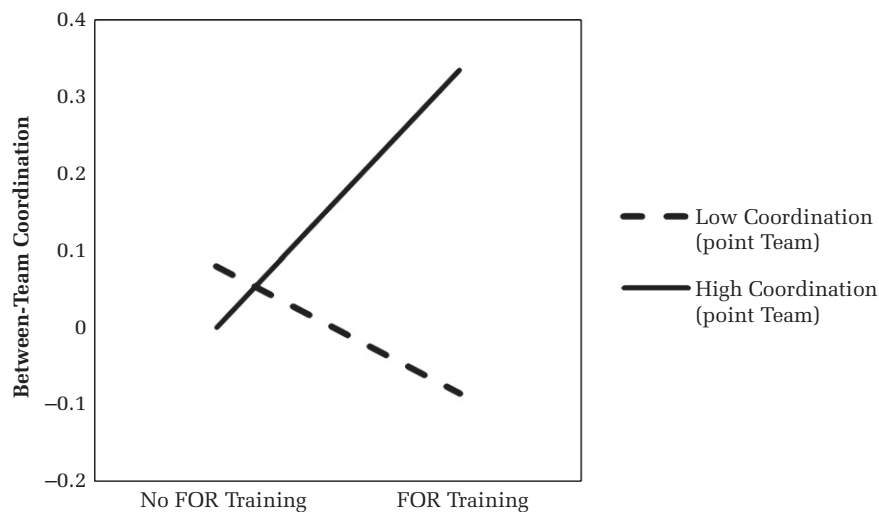
	Model 1	Model 2	Model 3
Frame-of-reference training	.08 [†]	.02	−.04
Within-team coordination (point team)		.20*	.15*
Within-team coordination (support team)		.49*	.49*
Between-team coordination			.18*
Total R^2	.01 [†]	.30*	.32*
ΔR^2		.29*	.02*

Notes: $n = 249$; observations are multiteam systems of 14 members each; standardized betas are reported.

* $p < .05$

[†] $p < .10$; one-tailed tests

FIGURE 2
Interaction of Within-Team Coordination (Point Team) and Frame-of-Reference Training on Between-Team Coordination

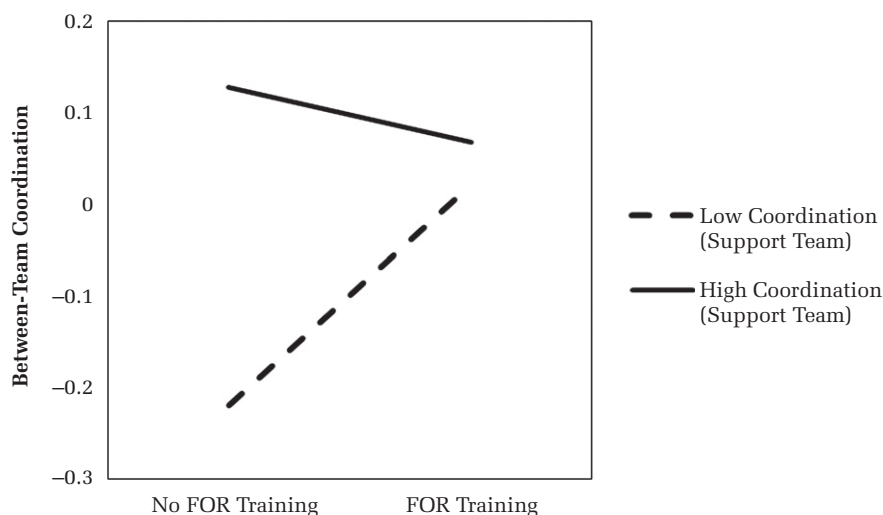


Note: FOR training = frame-of-reference training.

support teams and frame-of-reference training was also significant ($\beta = -.15$; $p < .05$; see Figure 3 for the form of this interaction, which uses values of ± 1 *SD* for within-team coordination). As expected, the nature of this interaction suggests that frame-of-reference training was most impactful for support

teams with low levels of within-team coordination. Overall, 11% of the variance in between-team coordination was explained by frame-of-reference training, within-team coordination, and the interactions between frame-of-reference training and within-team coordination.

FIGURE 3
Interaction of Within-Team Coordination (Support Team) and Frame-of-Reference Training on Between-Team Coordination



Note: FOR training = frame-of-reference training.

To fully test Hypotheses 3 and 4, we ran moderated mediation analyses in Mplus 6.1 (Muthén & Muthén, 2010), in line with recommendations provided by Preacher, Rucker, and Hays (2007) and Edwards and Lambert (2007). Using bootstrapping procedures, we obtained bias-corrected 95% confidence intervals for estimates of the conditional indirect effects of these relationships. As predicted, the indirect effect for low levels of within-team coordination in point teams was not significant ($B = 0.00$, $SE = .02$, $p > .10$), while high levels of within-team coordination in point teams was significant ($B = 0.07$, $SE = 0.03$, $p < .05$). The confidence interval for the indirect effect of low levels of within-team coordination for point teams included zero (-0.03 , $.04$), while for high levels of within-team coordination for point teams it excluded zero (0.03 , 0.14). Also as predicted, the indirect effect for low levels of within-team coordination in support teams was significant ($B = 0.05$, $SE = .02$, $p < .05$), while high levels of within-team coordination in support teams was marginally significant ($B = 0.03$, $SE = .02$, $p < .10$). The confidence interval for the indirect effect of low levels of within-team coordination for support teams excluded zero (0.01 , 0.10), while for high levels of within-team coordination for support teams it included zero (0.00 , 0.08). Thus, we found evidence that the indirect effects of frame-of-reference training on multiteam system performance via between-team coordination were strongest when (a) within-team coordination for point teams was high, as well as when (b) within-team coordination for support teams was low, supporting Hypotheses 3 and 4 (see Table 4).

As a supplementary analysis, we examined the interaction between within-team coordination for

point teams and within-team coordination for support teams. Given our focus on the moderation of our central mediated model, we did not hypothesize this interaction. However, if significant, this interaction would support our underlying logic that within-team coordination in both point and support teams benefits between-team coordination. Controlling for our independent variables and the other two interactions, the results were significant ($\beta = .13$; $p < .05$). The three-way interaction between both types of within-team coordination and frame-of-reference training on between-team coordination was not significant ($\beta = .03$; $p > .10$).

To further explore our results, we used a regression equation that predicted scores for between-team coordination from our three independent variables and all two-way interactions when entering high and low levels (i.e., ± 1 *SD*) of our independent variables. The predicted score for those with high levels of both types of within-team coordination and frame-of-reference training was 0.45 , corresponding to the 46th percentile of between-team coordination. The predicted score for those with high levels of both types of within-team coordination but no frame-of-reference training was -0.14 , corresponding to the 21st percentile of between-team coordination. The predicted score for those with low levels of both types of within-team coordination with frame-of-reference training was 0.27 , corresponding to the 35th percentile of between-team coordination. The predicted score for those with low levels of both types of within-team coordination and no frame-of-reference training was -0.32 , corresponding to the 19th percentile of between-team coordination. These results illustrate that training resulted in an increase of the percentile

TABLE 4
Moderated Mediation Simple Effects

Moderator variable	Effect		
	Direct	Indirect	Total
<i>Within-team coordination (point team)</i>			
Low	0.03	0.00	0.04
High	-0.02	0.07*	0.05*
Difference	0.05	0.07*	0.01*
<i>Within-team coordination (support team)</i>			
Low	0.01	0.05*	0.06*
High	-0.11*	0.03†	-0.08*
Difference	0.12*	0.02*	0.14*

Notes: $n = 249$; observations are multiteam systems of 14 members each; standardized betas are reported.

* $p < .05$

† $p < .10$

rank of between-team coordination regardless of whether both types of within-team coordination were high (a 25 percentile increase) or low (a 16 percentile increase).

Using the same regression equation, the predicted score for those with high levels of point team coordination but low levels of support team coordination in the presence of frame-of-reference training was 0.11, corresponding to the 27th percentile of between-team coordination. The predicted score for those with high levels of point team coordination but low levels of support team coordination with no frame-of-reference training was -0.48 , corresponding to the 24th percentile of between-team coordination. The predicted score for those with low levels of point team coordination but high levels of support team coordination in the presence of frame-of-reference training was 0.61, corresponding to the 57th percentile of between-team coordination. The predicted score for those with low levels of point team coordination but high levels of support team coordination with no frame-of-reference training was 0.01, corresponding to the 24th percentile of between-team coordination. These results suggest that, although the actions of point teams are most central to multiteam system actions, the overall system may be best off when well-coordinated support team actions can enhance the value of the presence of frame-of-reference training on between-team coordination, such that they may facilitate point teams' actions regardless of how well those point teams coordinate on their own.

To better understand how each of our predictor variables collectively contributed to multiteam system performance, we ran a hierarchical regression analysis. After controlling for frame-of-reference training, we found that within-team coordination for point teams ($\beta = .15$; $p < .05$), within-team coordination for support teams ($\beta = .49$; $p < .05$), and between-team coordination ($\beta = .18$; $p < .05$) each related uniquely and positively to multiteam system performance (see Table 3 for a summary of these results). Overall, 32% of the variance in multiteam system performance was explained by these predictors, with between-team coordination positively predicting multiteam system performance above and beyond within-team coordination ($\Delta R^2 = .02$, $p < .05$).

As some of these results were accompanied with relatively small effect sizes, we explored their practical significance by calculating an index of binomial effects size display (Rosenthal & Rubin,

1982). This analysis demonstrated that, when defining mission success as being in the top 10% of performers across both conditions (i.e., the top 25 of all multiteam systems), 15 of 25 of such teams were in the frame-of-reference training condition. This means that multiteam systems in the experimental condition were 1.5 times more likely than those in the control condition to succeed on the mission. At the other end of the continuum, when defining mission failure as being in the bottom 10% of performers in both conditions (i.e., the bottom 25 of all multiteam systems), 19 of 25 were in the untrained condition. This means that multiteam systems in the control condition were 3 times more likely than those in the experimental condition to experience mission failure. Given the importance of the typical missions performed in our focal organization (i.e., the USAF), these differences in success and failure rates are not trivial.

DISCUSSION

Building from information processing theory and extending representational gaps theorizing, we advanced an understanding of the critical role that between-team coordination plays in multiteam system performance. We showed that frame-of-reference training, which is aimed at reducing representational gaps, can enhance multiteam system performance by improving between-team coordination. We also showed that within-team coordination positively moderates the effect of frame-of-reference training on between-team coordination and multiteam system performance. For point teams, this *enhances* the main effects of frame-of-reference training on between-team coordination and multiteam system performance, while for support teams it *compensates* for the main effects. Thus, we demonstrated that processes from different types of specialized teams can differentially impact the pattern in which frame-of-reference training is beneficial to multiteam systems.

Theoretical Contributions

The primary theoretical contribution of this study is that we extended knowledge of what enables between-team coordination in multiteam systems. First, we introduced frame-of-reference training and within-team coordination as two important antecedents to between-team coordination. In doing so, we provide a meaningful extension to information processing theory, arguing that efforts to reduce

uncertainty from various sources may not just have additive effects, as originally suggested by Tushman and Nadler (1978), but that they may have interactive effects as well. Extending representational gaps theorizing, we elaborated on the role of distributed subgoals in creating representational gaps, which enriches the original theory to account for more complex goal hierarchies. By explicating how representational gaps may cause problems *between* teams in addition to *within* teams, we extended the theory to an additional level of analysis. This work provides a richer explanation for why multiteam systems must align component teams' efforts. In line with our theorizing, we focused specifically on the presence of subgoals that differ between teams within multiteam systems. However, accounting for subgoals in representational gaps theorizing likely has valuable implications when applied to the team level (e.g., it may help to better explain cognitive differences between subgroups in teams; cf. Carton & Cummings, 2012) or broader organizational level (e.g., it may help explain misalignments between departments).

We also extend current theorizing regarding the benefits of centralized processes in multiteam systems. Although centralized leadership functions such as between-team coordination may be well suited to multiteam systems (Davison et al., 2012; Lanaj et al., 2013), this alone is not necessarily sufficient to ensure that coordination processes result in strong multiteam system performance. Our theory suggests that the potential upsides of centralized between-team coordination (e.g., reduced informational complexity) can still be hampered by cognitive differences in how members represent and understand the problem space facing the multiteam system. We demonstrated that centralized coordination (i.e., between-team coordination enacted by a leadership team) can be enhanced by a simple manipulation targeted at reducing representational gaps within multiteam systems. That said, training does not decrease the need for centralized coordination functions. Our work helps shift the conversation away from which forms (e.g., centralized versus decentralized) and functions (e.g., coordination) of leadership are important to multiteam systems, and toward understanding how the fundamental processes (e.g., shared cognitive representations) that underlie leaders' efforts can be effective.

Our theory also expands the prevailing conceptualization of integrative functions from being predominantly focused on providing top-down

coordination (e.g., providing updates, information, timing cues, directions; DeChurch et al., 2011; DeChurch & Marks, 2006; Kozlowski et al., 1996) to facilitating bottom-up processes (Zaccaro & DeChurch, 2012). Our theory is in line with past work on the importance of multiteam system leadership overseeing between-team coordination. However, between-team coordination is also impacted by the quality of coordination processes *within* component teams—particularly when receiving frame-of-reference training. Our work thus suggests that between-team coordination (a system-level process) is impacted by efforts lower in the hierarchy (i.e., in teams). This is an important shift, as prior work has largely emphasized the independent effects of between-team coordination on multiteam system performance (e.g., DeChurch & Marks, 2006).

More practically, we outline a simple way to reduce representational gaps, which serves as an additional extension of that theory. Prior suggestions for reducing representational gaps included encouraging understanding, enhancing functional knowledge for non-experts, encouraging adaptation and insight, increasing motivation to learn, and reducing biases about other team members (Cronin & Weingart, 2007). Each of these approaches may be fruitful. However, our theorizing suggests that an additional approach may be to provide standardized scales upon which individuals may assess performance demands, providing a common language between and within component teams and their respective members.

Our findings also indicate that the pattern of effects of frame-of-reference training in multiteam systems depend on component teams' specific functions. This highlights the theoretical importance of accounting for functional differences when considering how team-level processes impact system-level processes. Practically speaking, without accounting for these differences, an important effect of frame-of-reference training would have been overlooked. That is, without disentangling the effects of within-team coordination by team function, the average interactive effect of within-team coordination (i.e., across team types) and frame of reference on between-team coordination appears to be a mild enhancement effect, which overlooks the theoretical and practical differences between teams relative to how their tasks contribute to system-level goal hierarchies. Although we expected a main effect of within-team coordination for support teams on between-team coordination (i.e., Hypothesis 2b),

our broader arguments and analyses may help to explain why this was not significant but the interaction with frame-of-reference training was (i.e., Hypothesis 4). Unless within-team coordination in support teams is particularly high, other sources of uncertainty must be reduced if the coordinated actions of support teams are to be integrated into and benefit the system. This suggests that information processing during between-team coordination can be taxed to the point that essential inputs are not incorporated unless important sources of uncertainty can be minimized.

This work also highlights that within-team coordination, which is broadly viewed as the *sine qua non* of effective teams, may not be sufficient to ensure coordination between teams in more complex systems. What is important for team effectiveness may change in situations where there are interdependencies between teams. Multiteam systems that are effective at within-team coordination may still fail due to their inability to meet between-team coordination requirements. It is plausible that other processes may demonstrate similar relationships.

Managerial Implications

The chief implication for managers is that a potential downside of multiteam complexity and specialization (i.e., idiosyncrasies between component teams that can make coordination difficult) can be mitigated by a simple training manipulation. Frame-of-reference training develops standardized confidence ratings as a common language for translating judgments that are based on idiosyncratic, functionally specific factors into a generically interpretable format easily understood by individuals who may not share the same area of expertise. An advantage of frame-of-reference training is its cost efficiency and ease of implementation. This is particularly valuable, in that traditional means of increasing coordination in teams (e.g., cross-training) would be difficult to implement between teams in multiteam systems.

Further, our findings emphasize the importance of reducing representational gaps in multiteam systems, which we acknowledge may be possible in ways other than just frame-of-reference training. Although this study was partially motivated by the fact that alternative methods for doing so are largely lacking, some other possibilities have been suggested that may provide managers with additional tools for developing effective leadership functions (e.g., Cronin & Weingart, 2007; DeChurch & Marks,

2006). We urge managers to focus on the ends rather than the means, such that they triangulate efforts to improve between-team coordination via reducing representational gaps through whatever means possible.

We also urge managers to account for team functions when implementing and assessing the effects of efforts to enhance between-team coordination. Extrapolating from our findings, decisions to implement interventions such as frame-of-reference training might benefit from assessing levels of coordination in each component team type, and considering how to balance this approach with other more traditional methods of enhancing team-level coordination when appropriate. For instance, if within-team coordination in point teams is low, it may be more effective to enhance that process before investing in frame-of-reference training. On the other hand, if within-team coordination is low in support teams, it may not be worth the effort to seek to enhance that process if the intent is to enhance between-team coordination and frame-of-reference training is an option. Of course, an advantage of frame-of-reference training is that, on average, it enhances between-team coordination regardless, and is relatively cheap and simple to implement. Our broader point is that managers should account for the complexities of multiteam systems before deciding on any given intervention.

Limitations and Future Research

The behavioral indicators we relied on served as objective measures of our constructs, but, as with any method, there are shortcomings to this approach. Specifically, the trade-off is that we did not capture more psychological factors that are likely important. For example, we did not formally test for the effect of frame-of-reference training on multiteam system cognitive structures (i.e., representational gaps or other forms of shared cognition), as representational gaps were a built-in constant of our research design. However, our findings regarding the effect of training on between-team coordination were guided by theory suggesting that shared cognition may be enhanced by directly targeting shared cognitive structures (cf. Cannon-Bowers, Salas, & Converse, 1993). This opens the possibility for other ways to reduce representational gaps. For example, related work suggests that interaction patterns developed during team (and, by extension, multiteam system) development may impact the presence of representational gaps

(Pearsall, Ellis, & Bell, 2010), which could be targeted by various interventions.

Additionally, experimentally introducing a moderating effect, such as frame-of-reference training, as we did can be a strong way to test for the presence of an expected mediating variable—such as cognitive structures or processes like representational gaps or shared mental models—which likely at least partially explains the linkages that between-team coordination has with both frame-of-reference training and within-team coordination (Spencer, Zanna, & Fong, 2005). Spencer et al. (2005) showed that this is particularly true when the mediating variable (e.g., representational gaps) is difficult to measure, as it was in our context, and when the manipulation has been demonstrated to impact the expected mediating variable (i.e., differences in cognition), as frame of reference has been in prior research. Thus, our design increases our confidence that the relationships in our model are due to the reasons provided in our theoretical rationale. Nevertheless, future work should investigate the specific cognitive and motivational mechanisms altered by frame-of-reference training in multiteam systems. It should also account for additional processes that may impact between-team coordination.

We carefully followed established guidelines in our training approach and believe the simplicity and streamlined nature of our manipulation highlights the potential of frame-of-reference training (i.e., more intensive training may have even more drastic effects). However, future work should explore alternate approaches to providing frame-of-reference training. For example, whereas we focused the training within the leadership team but included component teams in the more general aspects of it, one might alternatively direct training only to component teams, or equally across all multiteam system members. Varied training techniques (e.g., longer duration, more feedback-driven practice) might also have different effects, which should be explored. It might be valuable, too, to vary the size of representational gaps via cross-training or role rotation, investigating whether frame-of-reference training is especially important in contexts where representational gaps are larger versus smaller. It would also be important to examine if different levels of frame-of-reference training have different effects. For example, one could examine curvilinear effects associated with this intervention. Theoretically, if training time and cost is fixed, frame-of-reference training could start cutting into the training necessary to learn one's own

specialization. Future work should also examine other interventions that might enhance between-team coordination, whether by impacting shared cognition or other potential antecedents (e.g., relational commitment or anticipation of needs; Lawler & Yoon, 1996; Rico et al., 2008), or by addressing representational gaps that arise for very different reasons.

Our arguments suggest that shared cognition can enhance multiteam system performance in a complex and dynamic task. However, high levels of shared cognition in extremely dynamic environments may relate to rigid thinking that leads to maladaptive (e.g., inflexible) responses (e.g., Vashdi, Bamberger, & Erez, 2013). The extent to which shared cognition is a dynamic state that shifts over time as teams engage in learning processes in dynamic environments has generally been understudied. Although our results suggest that higher levels of shared cognition are likely to be beneficial across functionally differentiated units in dynamic contexts (cf. Faraj & Xiao, 2006; Majchrzak, More, & Faraj, 2012), the possibility that this may not always be the case, or that the utility of high levels of shared cognition might shift over time, warrants further investigation.

Many of our results were accompanied by relatively small effect sizes. We believe that the binomial effects size display that we report helps to explain why even these small effect sizes are practically important. Additionally, it is important to note that small effect sizes can have practical importance when (a) manipulations of the independent variable in question are minimal, and (b) when an independent variable operates in novel domains (Prentice & Miller, 1992). Our intervention was minimal, in that it consisted of a brief verbal explanation and a single one-page handout that together detailed some technical information and did not contain additional dimensions that would have directly impacted coordination or performance. We also examined a dependent variable that was distinct, in terms of both hierarchical level and complexity, from the typical outcomes of frame-of-reference training that are examined. Still, future work should examine additional interventions that may have even larger statistical and practical effects.

Lastly, we point out that, as with most research, one must extrapolate our findings beyond our specific research context with caution. This is particularly true given the breadth of multiteam system types, sizes, and goals (Mathieu et al., 2001). We had the advantage of a sample that consisted of working

professionals with real career trajectories at stake in a task that was amenable to laboratory-like control. Research that extends this work into additional settings (e.g., non-military, non-simulated tasks) may shed light as to how organizational cultures and structures influence the relationships we explored. We view our approach as a conservative test of our theoretical model, in that, if frame-of-reference training can benefit multiteam systems with moderate levels of coordination complexity (such as those in our sample), frame-of-reference training might be even more beneficial for teams with more complicated coordination demands. However, it is possible that our model will not hold in environments with particularly high levels of uncertainty or environmental dynamics, or where coordination between subgoals within multiteam systems is mundanely routine or non-existent. Thus, future work should examine what facilitates between-team coordination in different types of environments.

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APPENDIX A
TASK SAMPLE: SCREENSHOT OF COMMON OPERATIONAL PICTURE (COP)

Commander-Team Common Operational Picture

Player Game Grid Misc Test Printers Zoom

Turn 1 Initial Planning 0:00
Team: 0 SubTeam: 100 Player: 100

16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1

A B C D E F G H I J K L M N O P

16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1

A B C D E F G H I J K L M N O P

Ops Intel

1 2 3 4

Strike

Escort

Refuel

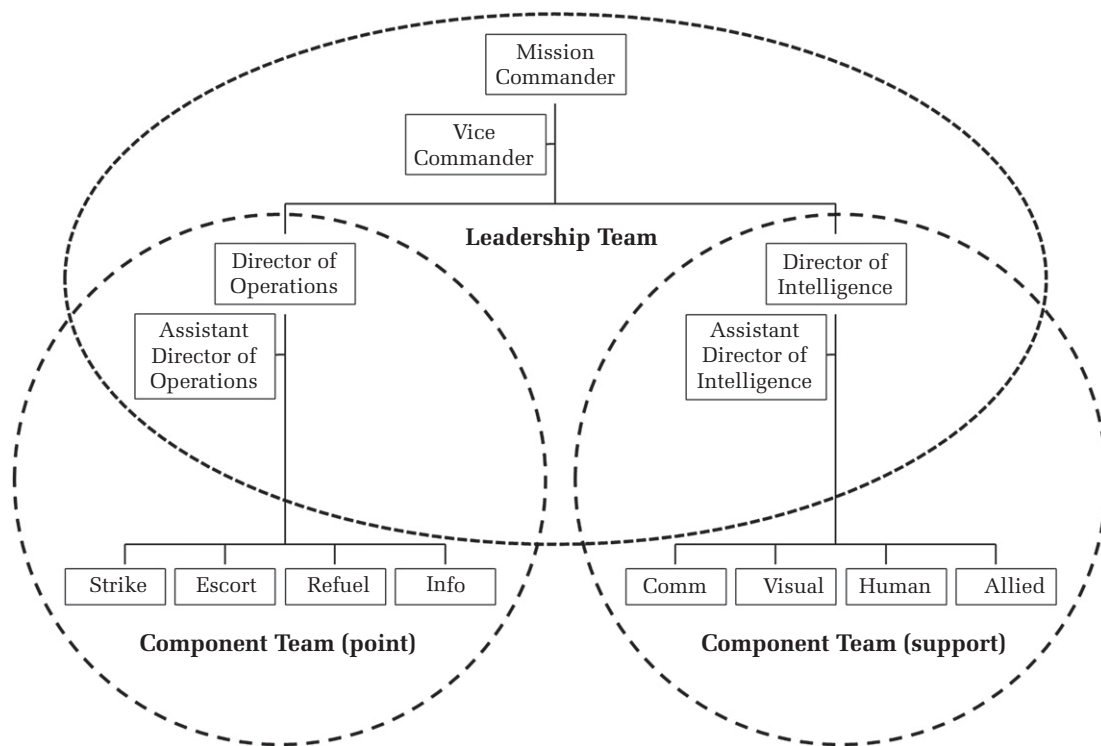
Info

☐ DA Debug 1000

StartGameStage A

	Opportunity		Threat		Unknown	
	Large	Small	Large	Small		
Fixed					W	X
Mobile					Y	Z

APPENDIX B
HIERARCHICAL STRUCTURE OF THE MULTITEAM SYSTEM



APPENDIX C

Confidence-Level Assignment Rules for Leadership Team

Level	Fixed Target Decision Rules	Mobile Target Decision Rules
0 (no rating)	Confidence not established—update ASAP	Confidence not established—update ASAP
1 (low)	Target detected once by one unvalidated intel unit	Target detected by one unvalidated intel unit OR Target detected by operational unit three or more rounds ago
2 (medium)	Target detected twice by one unvalidated intel unit ^a OR Target detected once by two different unvalidated intel units	Target detected by validated intel unit(s) two or more rounds ago OR Target detected by operational unit two or more rounds ago
3 (high)	Target detected three times or more by unvalidated intel unit(s) OR Target detected at least once by one validated intel unit	Target detected by validated intel unit(s) last turn but expected to move next turn OR Target detected by operational unit last turn but expected to move next turn
4 (ground truth)	Target detected by an operational unit	Target detected by validated intel unit last turn and you know the target will not move next turn OR Target detected by operational unit last turn and you know the target will not move next turn ^b

^a An “unvalidated intel [intelligence] unit” is one for which the “sweet spot” is yet to be determined. A “validated intel unit” is one for which the “sweet spot” has been established with 90% confidence.

^b Confidence ratings must be updated each round in order to accurately reflect up-to-date intelligence. In particular, do not forget to downgrade confidence estimates on mobile targets after each round.

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